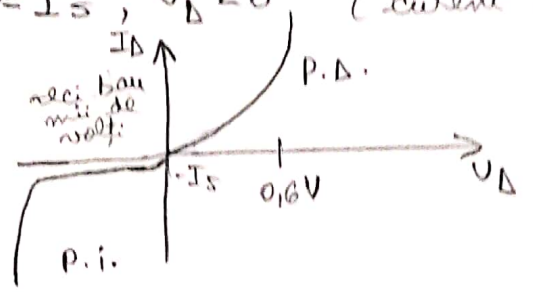
Pol. directă : $I_D = I_S\left(e^{\frac{U_D}{V_T}} - 1\right)$, $U_D > 0$ (curent de recombinare)

$V_T = \frac{kT}{q_0} \approx 26\,mV$

$I_S \ll 10^{-12}\,A$

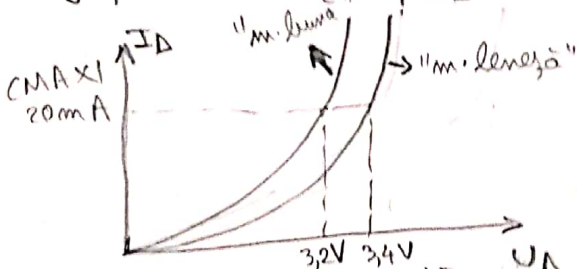
Pol. inv. : $I_D \approx I_S$, $U_D < 0$ (curent de generare)

Caracc. statică :



U_D (I_D ↓ diodă , ⇐ fotodiodă (P.i.; lumină → generare) I_D

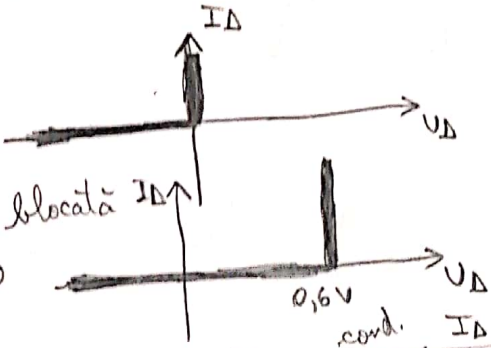
I_D ↑ L.E.D. (P.D.; recombinare $\varepsilon - G \rightarrow$ lumină)



M: 3% INC
15% L.E.D.

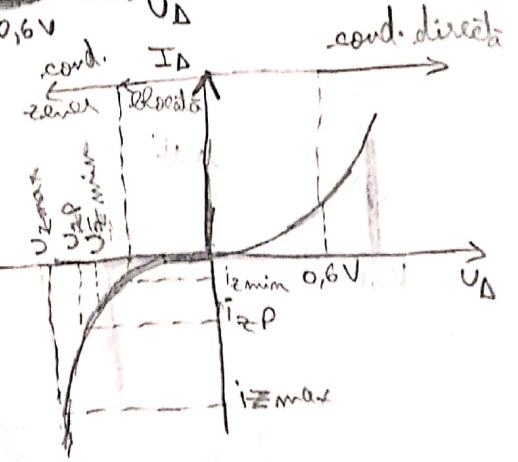
APROXIMĂRI MAT. PT. DIODE:

• DIODA IDEALĂ : $\begin{cases} I_D = 0 \,,\ U_D < 0 \\ U_D = 0 \,,\ I_D > 0 \end{cases}$



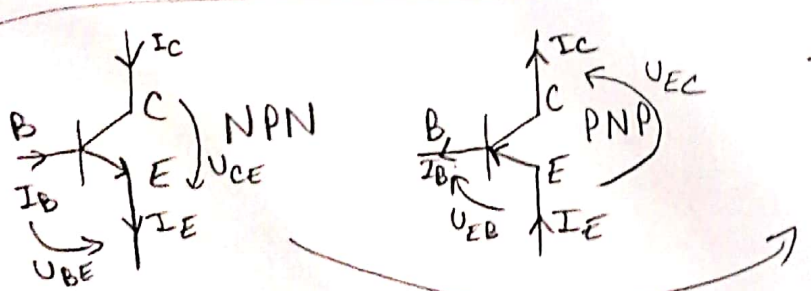
• TENS. CT. : $\begin{cases} I_D = 0 \,,\ U_D < 0{,}6\,V \text{ blocată} \\ U_D = 0{,}6\,V \,,\ I_D > 0 \text{ conducție} \end{cases}$

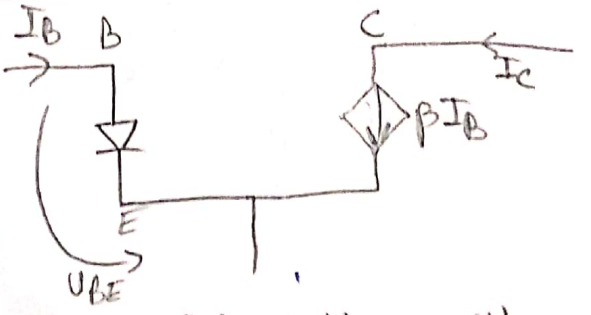
$I_D = I_S\left(e^{\frac{U_D}{V_T}} - 1\right) \approx I_S\, e^{\frac{U_D}{V_T}}$, $V_T \approx 25\,mV$



Dioda Zener : $\begin{cases} I_D > 0 \,,\ U_D = 0{,}6\,V & \text{conducție} \\ -U_{DZ} < U_D < 0{,}6\,V \,,\ I_D = 0 & \text{blocată} \\ I_D < 0 \,,\ U_D = -U_{DZ} & \text{cond. inv.} \end{cases}$

$I_C = C\frac{dU_C}{dt}$ ⇒ $I_C > 0$ U_C ↑ ; $I_C = 0$ U_C = ct ; $I_C < 0$ U_C ↓





$I_C = \beta I_B$

$I_E = I_C + I_B \approx I_C$ (RAN)

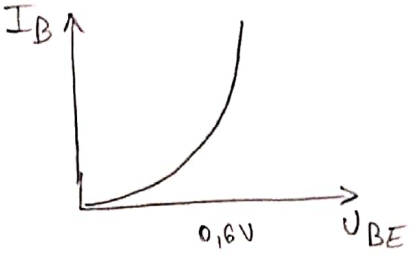
$U_{BE} \approx 0{,}6\,V$ $U_{CE} > 0{,}6\,V$

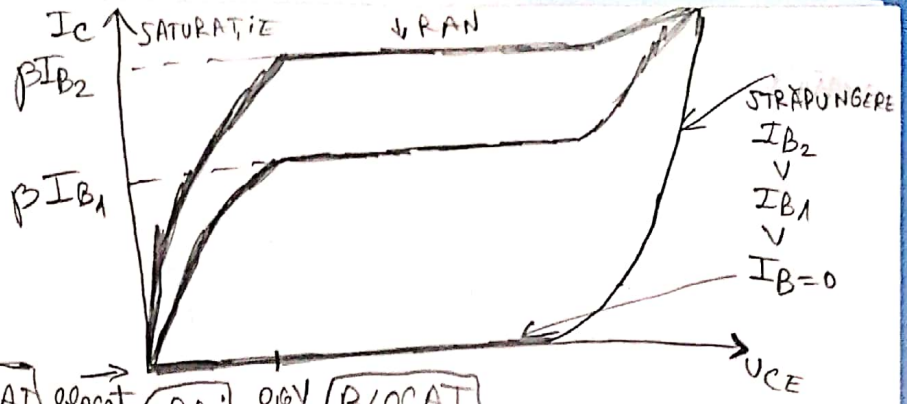
BE → P.D. → $U_{BE} = 0{,}6\,V$

CB → P.i. → $U_{CB} > 0$

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Caracteristica transfer:





	RAN		SAT (blocat)		RAI 0,6V		BLOCAT	
NPN	$U_{BE}>0$	$U_{CB}>0$	$U_{BE}>0$	$U_{CB}<0$	$U_{BE}<0$	$U_{CB}<0$	$U_{BE}<0$	$U_{CB}>0$
PNP	$U_{EB}>0$	$U_{BC}>0$	$U_{EB}<0$	$U_{BC}<0$	$U_{EB}<0$	$U_{BC}<0$	$U_{EB}<0$	$U_{BC}>0$

$I_C = \beta I_B$

$U_{BE} \simeq 0,6V$

β = foarte mare

$I_C = 0$

P.S.F. $\rightarrow I_C, U_{CE}$

$$R_{\text{DINAMICĂ DIODĂ}} = R_{DD} = \frac{dU_D}{dI_D} = \frac{V_T}{I_D} \approx \frac{1}{40 I_D}\ K\Omega$$

(I_D în mA)

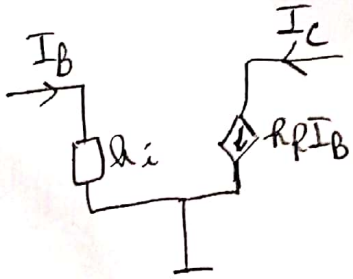
$f(E, u_i)$

1) P.S.F. $f(E,0) \Rightarrow U_0$ și R_{DD}
2) Liniarizăm circ. în jurul P.S.F. $f'(\,)$ liniar } aprox. semnal mic
3) $f'(0, u_i) \Rightarrow U_0(0, u_i)$
4) Superpoziție – efectul sumei e suma efectelor

$$f(E, u_i) = f(E,0) + f'(0, u_i) = U_0 + U_0(0, u_i)$$

în Reg. dinamic $\dashv\vdash$ devine —, la polarizare $\dashv\vdash \rightarrow$ — —

$$h_i = R_{DD} = \frac{1}{40 I_B} = \frac{\beta}{40 I_C}\ K, \quad h_f = \beta$$



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Amplificatoare:

$z_i = \frac{U_i}{I_i}$ (impedanța de intrare) $\quad I_{sc} = \frac{A_u U_i}{z_0}$ $\quad A_u = \frac{U_o}{U_i}$ (amplificare) $\quad z_0 = \frac{A_u U_i}{I_{sc}}$ (impedanța de ieșire)

	A_u	z_i	z_0
EC (emitor comun)	-200	2,5K	5K
CC (colector comun)	1	90K	25Ω

Ampl. op.: $V_{out} = A(V^+ - V^-)$

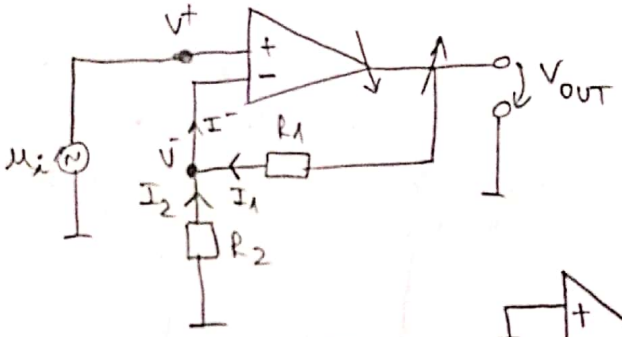
R.N. $\Rightarrow V^+ = V^-$

ideal: $\begin{cases} V_{OUT} = A(V^+ - V^-) \\ A \to \infty \quad (A = \infty) \\ I^+ = I^- = 0 \quad z_i = \infty \end{cases}$

Ampl. neinversor:

$V^+ = u_i$

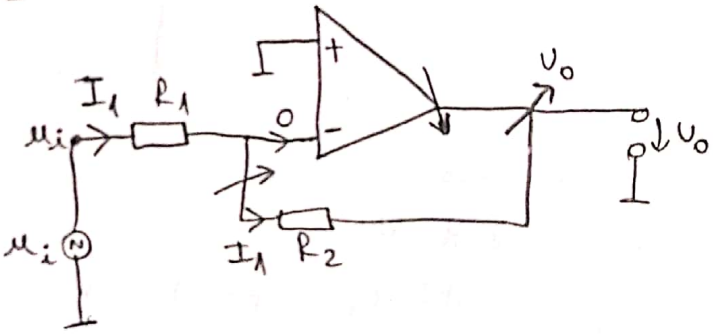
$V_{OUT} = u_i \frac{R_1 + R_2}{R_2}$



Ampl. op. inversor:

$V_{OUT} = -A \cdot \frac{u_i/R_1 + U_0/R_2}{\frac{1}{R_1} + \frac{1}{R_2}} =$

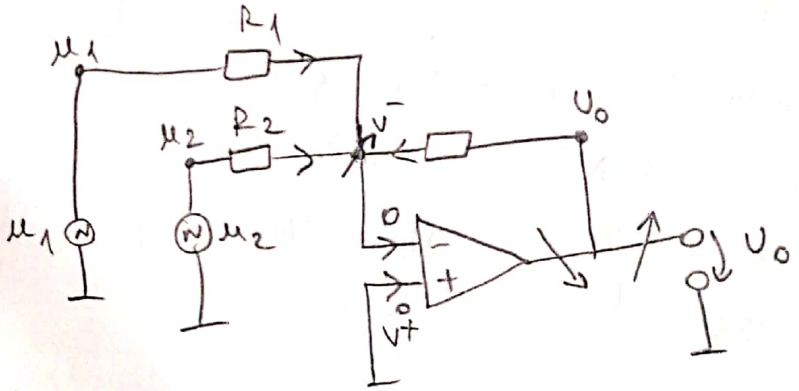
$= -A \cdot$



Ampl. op. sumator:

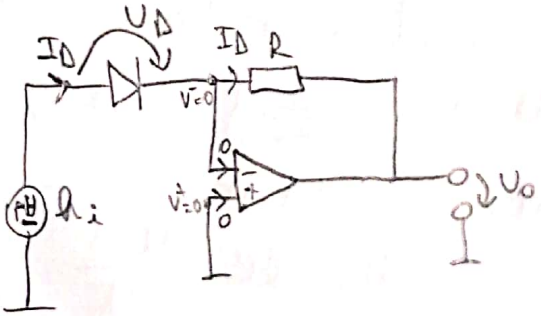
R.N. $\Rightarrow V^+ = V^- = 0$

$U_0 = -R_3\left(\frac{U_1}{R_1} + \frac{U_2}{R_2}\right)$



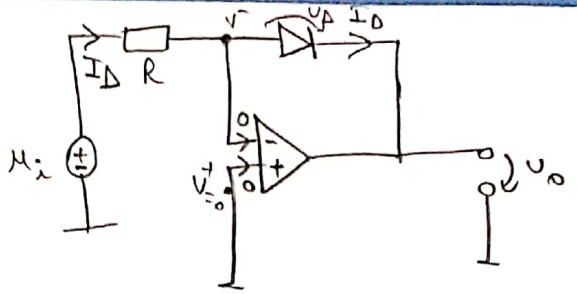
Ampl. op. exponențial:

$U_0 = R I_D = R I_S e^{\frac{U_D}{U_T}} \approx \ln(U_D)$



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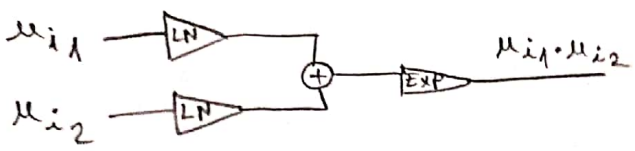
Ampl. op. logaritmic:



$v^+ = 0$ R.N. $\Rightarrow v^- = v^+ = 0$

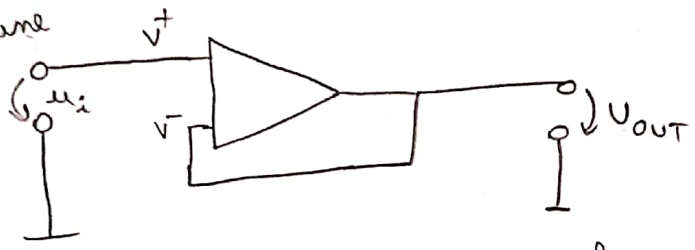
$U_0 = -V_T \ln \frac{u_i}{I_S R} \approx \ln u_i$

$u_{i1} \cdot u_{i2} = \exp(\ln u_{i1} + \ln u_{i2})$



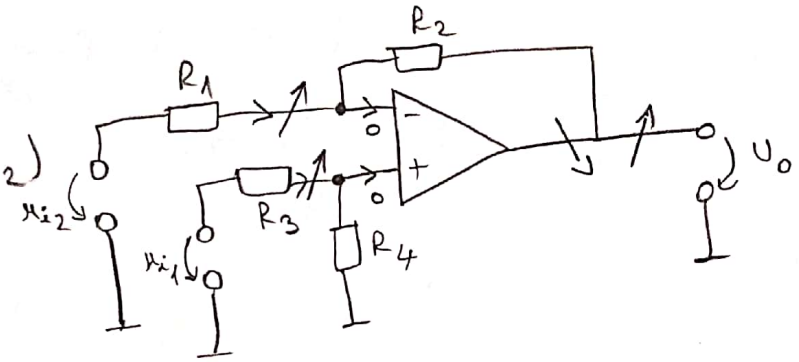
$\overset{+}{\ln}$ $\overset{-}{\exp}$

Repetor de tensiune:



$U_0 = u_i$

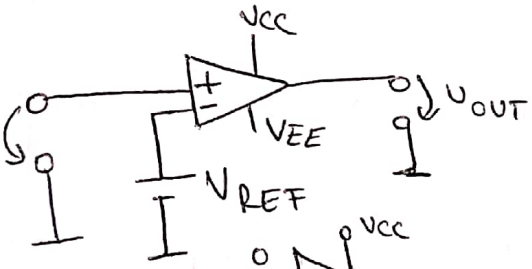
Ampl. diferențial:



$U_0 = K(u_{i1} - u_{i2})$

R.N. $\Rightarrow v^+ = v^-$

Comparator:

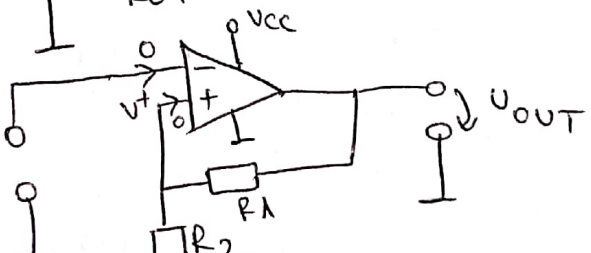


nu are R.N.

$U_{OUT} = A(u_i - V_{REF})$

$A \triangleq 10^6$

Comparator cu histerezis:

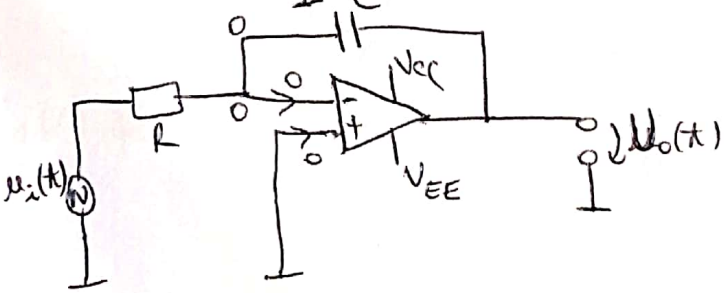


nu are R.N., are R.P.

are 2 stări $\begin{cases} U_{OUT} = V_{CC} \\ U_{OUT} = 0 \end{cases}$

$V_{EE} = 0V$

Circuit integrator



$U_0(t) = -\frac{1}{RC} \int u_i(t)\, dt$

pt. derivator $\dashv\vdash \rightarrow$ —▭—

$U_0(t) = -RC \frac{du_i(t)}{dt}$

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FTJ : $\frac{1}{1+SRC} = H(S)$

FTS : $\frac{SRC}{1+SRC} = H(S)$

$\omega = 2\pi f$

$H(S) = \frac{U_o(S)}{U_i(S)}$

$U_i(t) = \sin(\omega t)$

$U_i(t) \rightarrow U_i(S)$

$U_o(t) = |H(\omega j)| \sin(\omega t < H(j\omega))$

$R \rightarrow R$

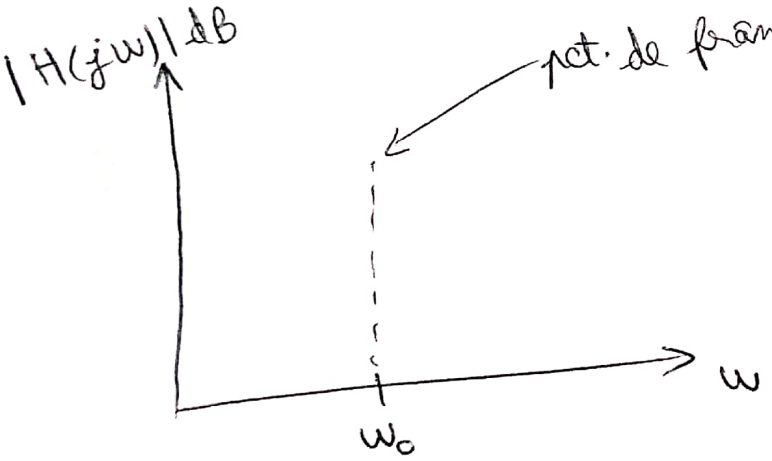
$C \rightarrow \frac{1}{SC}$

$L \rightarrow SL$

$A = |H(\omega j)|$

$dB = 20 \lg A$

$\omega_o = \frac{1}{RC}$



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